LAPLACE DISTRIBUTORS CORRESPOND BIJECTIVELY TO EXPONENTIAL MAPS ON THE MONOIDAL UNIT

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Editorial note. Source https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.FSCD.2024.9. © Marie Kerjean and Jean-Simon Pacaud Lemay, CC BY4.0. Statement and full author proof faithfully transcribed from the published PDF; original defining and proof diagrams retained.

Allowed standard prerequisites (editorial).

- Additive symmetric monoidal closed categories and monoidal differential coalgebra modalities. Publisher§2 and references[2],[18],[22]. Exact canonical-map equations(4)–(10) and the compatibility identities actually used in the proof are retained.

Background: Differential Categories

We follow the terminology and notation of the source. The underlying categorical structure is an additive symmetric monoidal category. We denote the underlying category by $\mathcal{C}$, the monoidal product by $\otimes$, the monoidal unit by I, the natural associativity isomorphism by $\alpha_{A,B,C} : A\otimes(B\otimes C) \to (A\otimes B)\otimes C$, the natural right unital isomorphism by $\rho_A : A \otimes I \to A$, and the natural symmetry isomorphism by $\sigma_{A,B} : A \otimes B \to B \otimes A$. Each homset is a commutative monoid with addition $+$ and zero 0, and composition and $\otimes$ are compatible with the additive structure.

A coalgebra modality is a comonad ! with digging $p_A :!A \to !!A$ and dereliction $d_A :!A \to A$, contraction $c_A :!A \to !A\otimes !A$ and weakening $w_A :!A \to I$, making each $!A$ a cocommutative comonoid and digging a comonoid morphism. A monoidal coalgebra modality furthermore has $\mu_{A,B} :!A\otimes !B \to !(A \otimes B)$ and $\mu_I : I \to !I$ making ! lax monoidal; p, d, c, w are monoidal transformations, and c, w are !-coalgebra morphisms. In the additive setting one obtains co-contraction $\bar{c}_A :!A\otimes !A \to !A$ and co-weakening $\bar{w}_A : I \to !A$, making each $!A$ a commutative monoid, and in fact a bimonoid. A monoidal differential modality additionally has a co-dereliction $\bar{d}_A : A \to !A$ with the differential-category axioms of source reference[2]. A differential linear category is an additive symmetric monoidal category with a monoidal differential modality. A differential linear closed category is one whose underlying symmetric monoidal category is closed.

We write $A \multimap B$ for internal homs and $\epsilon_{A,B} : (A \multimap B) \otimes A \to B$ for evaluation. For $f : C \otimes A \to B$, its unique curry $\lambda(f) : C \to A \multimap B$ satisfies

$$(\lambda(f) \otimes 1_A); \epsilon_{A,B} = f. \tag{4}$$

Composition is diagrammatic: $f; g$ denotes first f, then g. For $f : X \to A$ and $g : B \to Y$, the map $f \multimap g : A \multimap B \to X \multimap Y$ is characterized by

$$((f \multimap g) \otimes 1_X); \epsilon_{X,Y} = (1_{A\multimap B} \otimes f); \epsilon_{A,B}; g. \tag{5}$$

For a monoidal coalgebra modality the canonical $\xi_{A,B} :!(A \multimap B) \to !A \multimap !B$ satisfies

$$(\xi_{A,B} \otimes 1_{!A}); \epsilon_{!A,!B} = \mu_{A\multimap B,A}; !\epsilon_{A,B}. \tag{6}$$

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The dual object is $A^* = A \multimap I$; duality need not be involutive. The canonical isomorphism $\upsilon_I : I \to I^*$ is characterized by

$$(\upsilon_I \otimes 1_I); \epsilon_{I,I} = \rho_I, \qquad \upsilon_I^{-1} = \rho_{I^*}^{-1}; \epsilon_{I,I}. \tag{7}$$

The canonical $\nu_{A,B} : (A \otimes B)^* \to A \multimap B^*$ and its inverse satisfy

$$\begin{aligned} ((\nu_{A,B} \otimes 1_A) \otimes 1_B); (\epsilon_{A,B^*} \otimes 1_B); \epsilon_{B,I} &= \alpha^{-1}_{(A\otimes B)^*,A,B}; \epsilon_{A\otimes B,I}, \\ (\nu^{-1}_{A,B} \otimes 1_{A\otimes B}); \epsilon_{A\otimes B,I} &= \alpha_{A\multimap B^*,A,B}; (\epsilon_{A,B^*} \otimes 1_B); \epsilon_{B,I}. \end{aligned} \tag{8}$$

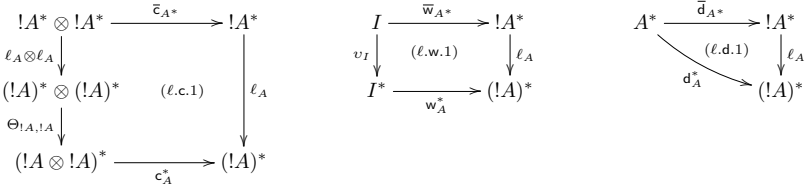
The canonical $\Theta_{A,B} : A^* \otimes B^* \to (A \otimes B)^*$ satisfies

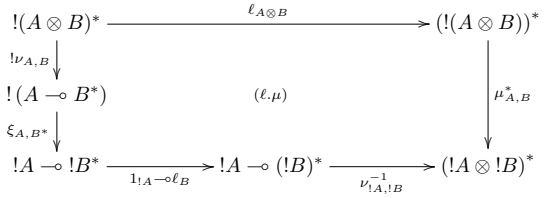
$$(\Theta_{A,B} \otimes 1_{A\otimes B}); \epsilon_{A\otimes B,I} = \tau_{A^*,B^*,A,B}; (\epsilon_{A,I} \otimes \epsilon_{B,I}); \rho_I, \tag{9}$$

where τ swaps the middle two terms. For $f : A \to B$, duality gives $f^* : B^* \to A^*$ characterized by

$$(f^* \otimes 1_A); \epsilon_{A,I} = (1_{B^*} \otimes f); \epsilon_{B,I}. \tag{10}$$

Definition1. For a differential linear closed category, a Laplace distributor is a natural transformation $\ell_A : !A^* \to (!A)^*$ such that the following original commutative diagrams hold:





These are the source axioms $(\ell.\bar{c}.1)$, $(\ell.\bar{w}.1)$, $(\ell.\bar{d}.1)$ and $(\ell.\mu)$.
Definition2. The original definition of the Laplace evaluator and its extranaturality equation are retained here, including the author's evaluator glyph:

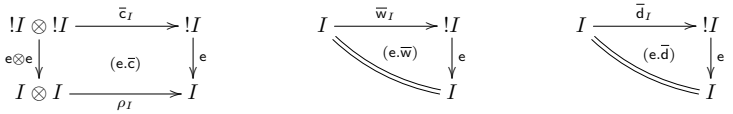
▶ **Definition 2.** *In a differential linear closed category with a Laplace distributor ℓ, the **Laplace evaluator** is the extranatural transformation $\eth_A : !A^* \otimes !A \to I$ defined as the composite:*

$$\eth_A :\; = \; !A^* \otimes !A \xrightarrow{\ell_A \otimes 1_{!A}} (!A)^* \otimes !A \xrightarrow{\epsilon_{!A,I}} I \tag{11}$$

By extranaturality, we mean that for all maps $f : A \to B$, the following equality holds:

$$(!f^* \otimes 1_A); \ell_A = (1_{!B^*} \otimes !f); \eth_B \tag{12}$$

Definition3. An I-exponential map is a map $e : !I \to I$ satisfying the original diagrams:



The author constructions

Proposition4. In a differential linear closed category with a Laplace distributor ℓ, define

$$e^\ell := !\upsilon_I; \ell_I; \mu_I^*; \upsilon_I^{-1} : !I \to I. \tag{13}$$

Proof. The key to this proof is using the part from the definition of a monoidal coalgebra modality[2, Definition1] which says that d, c, w are compatible with μ_I since they are monoidal transformations. So, by definition,

$$\mu_I; d_I = 1_I, \qquad \mu_I; w_I = 1_I, \qquad \mu_I; c_I = \rho_I^{-1}; (\mu_I \otimes \mu_I). \tag{14}$$

We need the dualized versions

$$d_I^*; \mu_I^* = 1_{I^*}, \quad w_I^*; \mu_I^* = 1_{I^*}, \quad c_I^*; \mu_I^* = (\mu_I \otimes \mu_I)^*; (\rho_I^{-1})^*. \tag{15}$$

First compute $(e.\bar{w})$:

$$\begin{aligned} \bar{w}_I; e^\ell &\overset{13}{=} \bar{w}_I; !\upsilon_I; \ell_I; \mu_I^*; \upsilon_I^{-1} \\ &\overset{nat.}{=} \bar{w}_{I^*}; \ell_I; \mu_I^*; \upsilon_I^{-1} \\ &\overset{\ell.\bar{w}.1}{=} \upsilon_I; w_I^*; \mu_I^*; \upsilon_I^{-1} \overset{15}{=} \upsilon_I; \upsilon_I^{-1} = 1_I. \end{aligned}$$

Next compute $(e.\bar{d})$:

$$\begin{aligned} \bar{d}_I; e^\ell &\overset{13}{=} \bar{d}_I; !\upsilon_I; \ell_I; \mu_I^*; \upsilon_I^{-1} \\ &\overset{nat.}{=} \upsilon_I; \bar{d}_{I^*}; \ell_I; \mu_I^*; \upsilon_I^{-1} \\ &\overset{\ell.\bar{d}.1}{=} \upsilon_I; d_I^*; \mu_I^*; \upsilon_I^{-1} \overset{15}{=} \upsilon_I; \upsilon_I^{-1} = 1_I. \end{aligned}$$

Using naturality of Θ and the following identity, which the authors leave to the reader to check, we compute $(e.\bar{c})$:

$$\Theta_{I,I}; (\rho_I^{-1})^*; \upsilon_I^{-1} = (\upsilon_I^{-1} \otimes \upsilon_I^{-1}); \rho_I. \tag{16}$$

$$\begin{aligned} \bar{c}_I; e^\ell &\overset{13}{=} \bar{c}_I; !\upsilon_I; \ell_I; \mu_I^*; \upsilon_I^{-1} \\ &\overset{nat.}{=} (!\upsilon_I \otimes !\upsilon_I); \bar{c}_{I^*}; \ell_I; \mu_I^*; \upsilon_I^{-1} \\ &\overset{\ell.\bar{c}.1}{=} (!\upsilon_I \otimes !\upsilon_I); (\ell_I \otimes \ell_I); \Theta_{!I,!I}; c_I^*; \mu_I^*; \upsilon_I^{-1} \\ &\overset{15}{=} (!\upsilon_I \otimes !\upsilon_I); (\ell_I \otimes \ell_I); \Theta_{!I,!I}; (\mu_I \otimes \mu_I)^*; (\rho_I^{-1})^*; \upsilon_I^{-1} \\ &\overset{nat.}{=} (!\upsilon_I \otimes !\upsilon_I); (\ell_I \otimes \ell_I); (\mu_I^* \otimes \mu_I^*); \Theta_{I,I}; (\rho_I^{-1})^*; \upsilon_I^{-1} \\ &\overset{16}{=} (!\upsilon_I \otimes !\upsilon_I); (\ell_I \otimes \ell_I); (\mu_I^* \otimes \mu_I^*); (\upsilon_I^{-1} \otimes \upsilon_I^{-1}); \rho_I \\ &\overset{13}{=} (e^\ell \otimes e^\ell); \rho_I. \end{aligned}$$

Therefore e^ℓ is an I-exponential map. The authors note that $(\ell.\mu)$ was not needed for this direction.

Proposition5. If $e : !I \to I$ is an I-exponential map, define $\ell_A^e : !A^* \to (!A)^*$ as the curry of

$$\mu_{A^*,A}; !\epsilon_{A,I}; e : !A^* \otimes !A \to I. \tag{17}$$

In other words, it is the unique map such that

$$(\ell_A^e \otimes 1_{!A}); \epsilon_{!A,I} = \mu_{A^*,A}; !\epsilon_{A,I}; e. \tag{18}$$

Then ℓ^e is a Laplace distributor; its induced evaluator is(17).

Proof. For readability the authors omit subscripts. The key is that evaluation is monic in its first argument: $(f \otimes 1); \epsilon = (g \otimes 1); \epsilon$ implies $f = g$. First, naturality:

$$\begin{aligned}(!f^* \otimes 1); (\ell^e \otimes 1); \epsilon &\overset{18}{=} (!f^* \otimes 1); \mu_{A^*,A}; \epsilon; e \\ &\overset{nat.}{=} \mu; !(f^* \otimes 1); !\epsilon_{A,I}; e \\ &\overset{10}{=} \mu; !(1 \otimes f); !\epsilon; e \\ &\overset{nat.}{=} (1 \otimes !f); \mu; !\epsilon; e \overset{18}{=} (1 \otimes !f); (\ell^e \otimes 1); \epsilon \\ &= (\ell^e \otimes 1); (1 \otimes !f); \epsilon \overset{10}{=} (\ell^e \otimes 1); ((!f)^* \otimes 1); \epsilon.\end{aligned}$$

Thus $!f^*; \ell^e = \ell^e; (!f)^*$. The co-structural compatibility relations from[2, Propositions2 and5] are

$$\begin{aligned}(\bar{w} \otimes 1); \mu &= \rho; w; \bar{w}, \\ (\bar{d} \otimes 1); \mu &= (1 \otimes d); \bar{d}, \\ (\bar{c} \otimes 1); \mu &= (1 \otimes 1 \otimes c); \tau; (\mu \otimes \mu); \bar{c}.\end{aligned} \tag{19}$$

For $(\ell.\bar{w}.1)$:

$$\begin{aligned}(\bar{w} \otimes 1); (\ell^e \otimes 1); \epsilon &\overset{18}{=} (\bar{w} \otimes 1); \mu; !\epsilon; e \overset{19}{=} \rho; w; \bar{w}; !\epsilon; e \\ &\overset{nat.}{=} \rho; w; \bar{w}; e \overset{e.\bar{w}}{=} \rho; w_A \overset{nat.}{=} (1 \otimes w); \rho \\ &\overset{7}{=} (1 \otimes w); (\upsilon \otimes 1); \epsilon = (\upsilon \otimes 1); (1 \otimes w); \epsilon \\ &\overset{10}{=} (\upsilon \otimes 1); (w^* \otimes 1); \epsilon.\end{aligned}$$

Thus $\bar{w}_{A^*}; \ell^e_A = \upsilon_I; w^*_A$. For $(\ell.\bar{d}.1)$:

$$\begin{aligned}(\bar{d} \otimes 1); (\ell^e \otimes 1); \epsilon &\overset{18}{=} (\bar{d} \otimes 1); \mu; !\epsilon; e \overset{19}{=} (1 \otimes d); \bar{d}; !\epsilon; e \\ &\overset{nat.}{=} (1 \otimes d); \epsilon; \bar{d}; e \overset{e.\bar{d}}{=} (1 \otimes d); \epsilon \overset{10}{=} (d^* \otimes 1); \epsilon.\end{aligned}$$

Thus $\bar{d}_{A^*}; \ell^e_A = d^*_A$. For $(\ell.\bar{c}.1)$:

$$\begin{aligned}(\bar{c} \otimes 1); (\ell^e \otimes 1); \epsilon_{!A,I} &\overset{18}{=} (\bar{c} \otimes 1); \mu; !\epsilon; e \\ &\overset{19}{=} (1 \otimes c); \tau; (\mu \otimes \mu); \bar{c}; !\epsilon; e \\ &\overset{nat.}{=} (1 \otimes c); \tau; (\mu \otimes \mu); (!\epsilon \otimes !\epsilon); \bar{c}; e \\ &\overset{e.\bar{c}}{=} (1 \otimes c); \tau; (\mu \otimes \mu); (!\epsilon \otimes !\epsilon); (e \otimes e) \\ &\overset{18}{=} (1 \otimes c); \tau; ((\ell^e \otimes 1) \otimes (\ell^e \otimes 1)); (\epsilon \otimes \epsilon) \\ &\overset{nat.}{=} (1 \otimes c); ((\ell^e \otimes \ell^e) \otimes 1); \tau; (\epsilon \otimes \epsilon) \\ &\overset{9}{=} ((\ell^e \otimes \ell^e) \otimes 1); (\Theta \otimes 1); (1 \otimes c); \epsilon \\ &\overset{10}{=} ((\ell^e \otimes \ell^e) \otimes 1); (\Theta \otimes 1); (c^* \otimes 1); \epsilon.\end{aligned}$$

Thus $\bar{c}_{A^*}; \ell^e_A = (\ell^e_A \otimes \ell^e_A); \Theta_{!A,!A}; c^*_A$. Lastly, the monoidal associativity axiom is

$$\alpha; (\mu \otimes 1); \mu = (1 \otimes \mu); \mu; !\alpha. \tag{20}$$

For $(\ell.\mu)$:

$$
\begin{aligned}
&(!\nu \otimes 1); (\xi \otimes 1); ((1 \multimap \ell^e) \otimes 1); (\nu^{-1} \otimes 1); \epsilon \\
&\overset{8}{=} (!\nu \otimes 1); (\xi \otimes 1); ((1 \multimap \ell^e) \otimes 1); \alpha; (\epsilon \otimes 1); \epsilon \\
&\overset{nat.}{=} \alpha; ((!\nu \otimes 1) \otimes 1); ((\xi \otimes 1) \otimes 1); (((1 \multimap \ell^e) \otimes 1) \otimes 1); (\epsilon \otimes 1); \epsilon \\
&\overset{5}{=} \alpha; ((!\nu \otimes 1) \otimes 1); ((\xi \otimes 1) \otimes 1); (\epsilon \otimes 1); \ell^e; \epsilon \\
&\overset{8}{=} \alpha; ((!\nu \otimes 1) \otimes 1); (\mu \otimes 1); (!\epsilon \otimes 1); \ell^e; \epsilon \\
&\overset{nat.}{=} \alpha; (\mu \otimes 1); (!(\nu \otimes 1) \otimes 1); (!\epsilon \otimes 1); \ell^e; \epsilon \\
&\overset{18}{=} \alpha; (\mu \otimes 1); (!(\nu \otimes 1) \otimes 1); (!\epsilon \otimes 1); \mu; !\epsilon; e \\
&\overset{nat.}{=} \alpha; (\mu \otimes 1); \mu; !((\nu \otimes 1) \otimes 1)); !(\epsilon \otimes 1); !\epsilon; e \\
&\overset{20}{=} (1 \otimes \mu); \mu; !\alpha; !((\nu \otimes 1) \otimes 1)); !(\epsilon \otimes 1); !\epsilon; e \\
&\overset{8}{=} (1 \otimes \mu); \mu; !\epsilon; e \overset{18}{=} (1 \otimes \mu); (\ell^e \otimes 1); \epsilon \\
&= (\ell^e \otimes 1); (1 \otimes \mu); \epsilon \overset{10}{=} (\ell^e \otimes 1); (\mu^* \otimes 1); \epsilon.
\end{aligned}
$$

Thus $!\nu_{A,B}; \xi_{A,B^*}; (1_{!A} \multimap \ell^e_B); \nu^{-1}_{!A,!B} = \ell^e_{A\otimes B}; \mu^*_{A,B}$. Therefore ℓ^e is a Laplace distributor, and(17) is its induced evaluator.

Theorem6

For a differential linear closed category, there is a bijective correspondence between Laplace distributors and I-exponential maps.

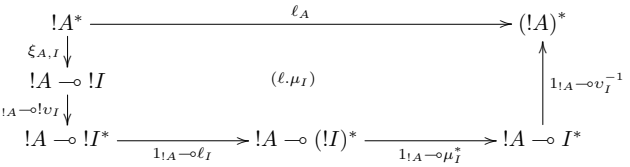
Proof. To show the constructions of Propositions4 and5 are inverses, we must show $e^{\ell^e} = e$ and $\ell^{e^\ell} = \ell$. The monoidal unital axiom is

$$(1_{!A} \otimes \mu_I); \mu_{A,I} = \rho_{!A}; !(\rho_A^{-1}). \tag{21}$$

First:

$$
\begin{aligned}
(e^{\ell^e} \otimes 1_I); (\upsilon_I \otimes 1); \epsilon_{I,I} &\overset{13}{=} (!\upsilon_I \otimes 1_I); (\ell^e_I \otimes 1_I); (\mu^*_I \otimes 1_I); (\upsilon_I^{-1} \otimes 1_I); (\upsilon_I \otimes 1); \epsilon_{I,I} \\
&= (!\upsilon_I \otimes 1_I); (\ell^e_I \otimes 1_I); (\mu^*_I \otimes 1_I); \epsilon_{I,I} \\
&\overset{10}{=} (!\upsilon_I \otimes 1_I); (\ell^e_I \otimes 1_I); (1_{(!I)^*} \otimes \mu_I); \epsilon_{!I,I} \\
&= (!\upsilon_I \otimes 1_I); (1_{!I^*} \otimes \mu_I); (\ell^e_I \otimes 1_{!I}); \epsilon_{!I,I} \\
&\overset{18}{=} (!\upsilon_I \otimes 1_I); (1_{!I^*} \otimes \mu_I); \mu_{I^*,I}; !\epsilon_{I^*,I}; e \\
&\overset{21}{=} (!\upsilon_I \otimes 1_I); \rho_{!I^*}; !(\rho^{-1}_{I^*}); !\epsilon_{I^*,I}; e \\
&\overset{nat.}{=} \rho_{!I}; !\upsilon_I; !(\rho^{-1}_{I^*}); !\epsilon_{I^*,I}; e \\
&\overset{7}{=} \rho_{!I}; !\upsilon_I; !\upsilon_I^{-1}; e = \rho_{!I}; e \\
&\overset{nat.}{=} (e \otimes 1_I); \rho_I \overset{7}{=} (e \otimes 1_I); (\upsilon_I \otimes 1); \epsilon_{I,I}.
\end{aligned}
$$

So $e^{\ell^e}; \upsilon_I = e; \upsilon_I$, and since υ_I is an isomorphism, $e^{\ell^e} = e$. On the other hand, the authors leave it as an exercise to check that taking $B = I$ in $(\ell.\mu)$ gives the following commuting diagram:



From this:

$$\begin{aligned}(\ell_A^{e^\ell} \otimes 1_{!A}); \epsilon_{!A,I} &\overset{18}{=} \mu_{A^*,A}; !\epsilon_{A,I}; e^\ell \\ &\overset{13}{=} \mu_{A^*,A}; !\epsilon_{A,I}; !\upsilon_I; \ell_I; \mu_I^*; \upsilon_I^{-1} \\ &\overset{6}{=} (\xi_{A,I} \otimes 1_{!A}); \epsilon_{!A,!I}; !\upsilon_I; \ell_I; \mu_I^*; \upsilon_I^{-1} \\ &\overset{5}{=} (\xi_{A,I} \otimes 1_{!A}); ((1_{!A} \multimap !\upsilon_I) \otimes 1_{!A}); ((1_{!A} \multimap \ell_I) \otimes 1_{!A}); \\ &\qquad ((1_{!A} \multimap !\upsilon_I) \otimes \mu_I^*); ((1_{!A} \multimap \upsilon_I^{-1}) \otimes 1_{!A}); \epsilon_{!A,I} \\ &\overset{\ell.\mu_I}{=} (\ell_A \otimes 1_{!A}); \epsilon_{!A,I}.\end{aligned}$$

So $\ell_A^{e^\ell} = \ell_A$. ◀

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